TUGAS TERBIMBING
PENGENALAN LIBRARY NETWORKX

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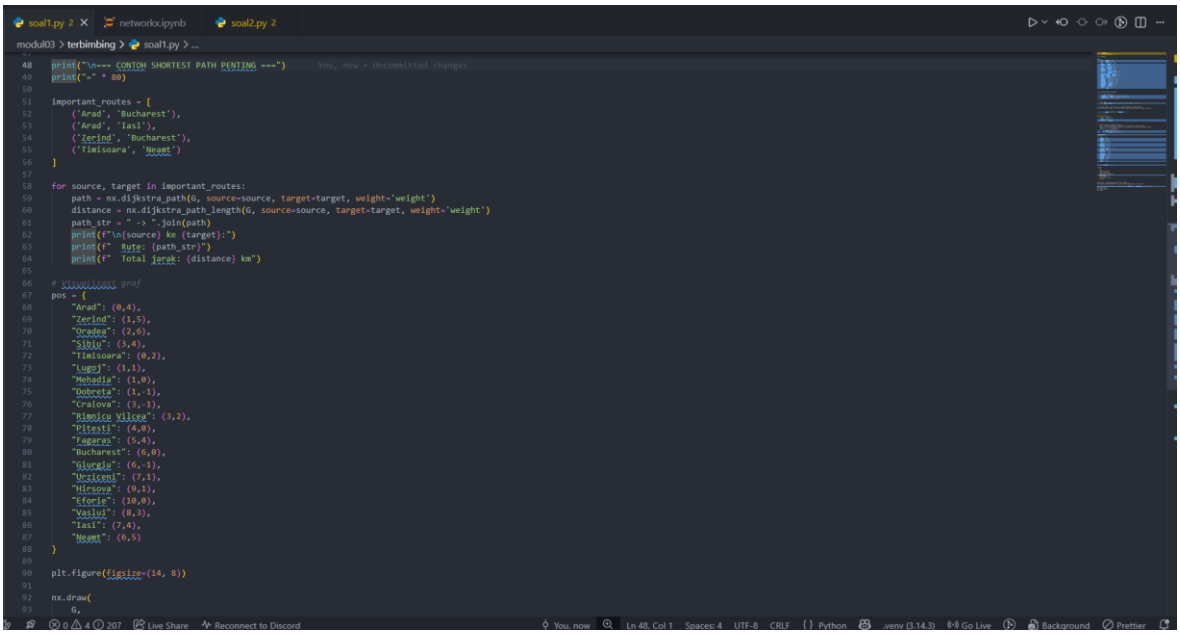
Soal 1



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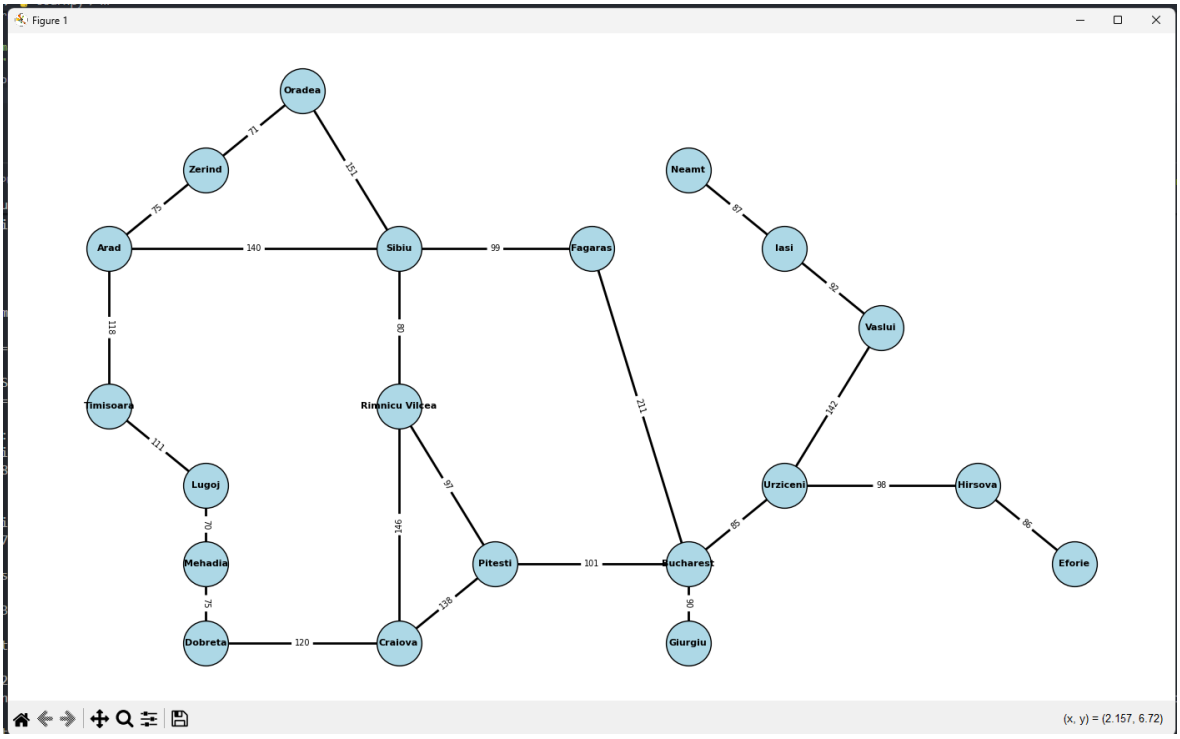
```python
pos = {
    "Arad": (0,4),
    "Zerind": (1,5),
    "Oradea": (2,6),
    "Sibiu": (3,4),
    "Timisoara": (0,2),
    "Lugoj": (1,1),
    "Mehadia": (1,0),
    "Dobreta": (1,-1),
    "Craiova": (3,-1),
    "Rimnicu Vilcea": (3,2),
    "Pitesti": (4,0),
    "Fagaras": (5,4),
    "Bucharest": (6,0),
    "Giurgiu": (6,-1),
    "Urziceni": (7,1),
    "Hirsova": (9,1),
    "Eforie": (10,0),
    "Vaslui": (8,3),
    "Iasi": (7,4),
    "Neamt": (6,5)
}

plt.figure(figsize=(14, 8))

nx.draw(
    G,
    pos,
    with_labels=True,
    node_size=1500,
    node_color="lightblue",
    edgecolors="black",
    font_size=8,
    font_weight="bold",
    width=2
)

edge_labels = nx.get_edge_attributes(G, 'weight')
nx.draw_networkx_edge_labels(G, pos, edge_labels=edge_labels, font_size=7)

plt.title("Romania Map Graph - Weighted Graph", fontsize=14, fontweight='bold', pad=20)
plt.axis("off")
plt.tight_layout()
plt.show()
```




```
Arad ke Bucharest:
  Rute: Arad -> Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest
  Total jarak: 418 km

Arad ke Iasi:
  Rute: Arad -> Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest -> Urziceni -> Vaslui -> Iasi
  Total jarak: 737 km

Zerind ke Bucharest:
  Rute: Zerind -> Arad -> Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest
  Total jarak: 493 km

Timisoara ke Neamt:
  Rute: Timisoara -> Arad -> Sibiu -> Rimnicu Vilcea -> Pitesti -> Bucharest -> Urziceni -> Vaslui -> Iasi -> Neamt
  Total jarak: 942 km
```

Soal 2

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You, 1 minute ago | 1 author (You)

```python
import networkx as nx
import matplotlib.pyplot as plt

G_jawa = nx.Graph()

edges_jawa = [
    ('Jakarta', 'Bandung', 270),
    ('Jakarta', 'Cirebon', 327),
    ('Bandung', 'Cirebon', 120),
    ('Bandung', 'Yogyakarta', 373),
    ('Cirebon', 'Yogyakarta', 210),
    ('Cirebon', 'Semarang', 305),
    ('Yogyakarta', 'Semarang', 109),
    ('Yogyakarta', 'Surakarta', 60),
    ('Semarang', 'Surakarta', 97),
    ('Semarang', 'Surabaya', 369),
    ('Surakarta', 'Surabaya', 370),
    ('Surabaya', 'Malang', 94)
]

G_jawa.add_weighted_edges_from(edges_jawa)

print("=== GRAF KOTA-KOTA JAWA ===")
print(f"Jumlah node: {G_jawa.number_of_nodes()}")
print(f"Jumlah edge: {G_jawa.number_of_edges()}")
print()

print("=== SHORTEST PATH ===")
print(f"Jakarta ke Surabaya: {nx.shortest_path(G_jawa, 'Jakarta', 'Surabaya', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Jakarta', 'Surabaya', weight='weight')} km")
print()

print(f"Bandung ke Malang: {nx.shortest_path(G_jawa, 'Bandung', 'Malang', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Bandung', 'Malang', weight='weight')} km")
print()

print(f"Jakarta ke Yogyakarta: {nx.shortest_path(G_jawa, 'Jakarta', 'Yogyakarta', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Jakarta', 'Yogyakarta', weight='weight')} km")

# Posisi manual untuk menyusun gambar
pos = {
    'Jakarta': (0, 4),
    'Bandung': (2, 2),
    'Cirebon': (3, 4),
    'Yogyakarta': (6, 0),
    'Surakarta': (8, 1),
```

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```python
    ('Semarang', 'Surabaya', 369),
    ('Surakarta', 'Surabaya', 370),
    ('Surabaya', 'Malang', 94)
]

G_jawa.add_weighted_edges_from(edges_jawa)

print("=== GRAF KOTA-KOTA JAWA ===")
print(f"Jumlah node: {G_jawa.number_of_nodes()}")
print(f"Jumlah edge: {G_jawa.number_of_edges()}")
print()

print("=== SHORTEST PATH ===")
print(f"Jakarta ke Surabaya: {nx.shortest_path(G_jawa, 'Jakarta', 'Surabaya', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Jakarta', 'Surabaya', weight='weight')} km")
print()

print(f"Bandung ke Malang: {nx.shortest_path(G_jawa, 'Bandung', 'Malang', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Bandung', 'Malang', weight='weight')} km")
print()

print(f"Jakarta ke Yogyakarta: {nx.shortest_path(G_jawa, 'Jakarta', 'Yogyakarta', weight='weight')}")
print(f"Jarak: {nx.shortest_path_length(G_jawa, 'Jakarta', 'Yogyakarta', weight='weight')} km")

# Posisi manual untuk menyusun gambar
pos = {
    'Jakarta': (0, 4),
    'Bandung': (2, 2),
    'Cirebon': (3, 4),
    'Yogyakarta': (6, 0),
    'Surakarta': (8, 1),
    'Semarang': (7, 4),
    'Surabaya': (11, 4),
    'Malang': (10, 0)
}

plt.figure(figsize=(12, 6))
nx.draw(G_jawa, pos, with_labels=True, node_color='lightgreen',
        node_size=2000, font_size=9, font_weight='bold',
        edge_color='gray', width=2)
edge_labels = nx.get_edge_attributes(G_jawa, 'weight')
nx.draw_networkx_edge_labels(G_jawa, pos, edge_labels=edge_labels, font_size=8)
plt.title('Graf Kota-Kota Besar di Pulau Jawa')
plt.axis('off')
plt.tight_layout()
plt.show()
```

```
PS C:\Storage\Documents\Telkom University\Semester 4\Dasar Kecerdasan Artifisial\praktikum\modul03\terbimbing> python soal2.py
=== GRAF KOTA-KOTA JAWA ===
Jumlah node: 8
Jumlah edge: 12

=== SHORTEST PATH ===
Jakarta ke Surabaya: ['Jakarta', 'Cirebon', 'Yogyakarta', 'Surakarta', 'Surabaya']
Jarak: 967 km

Bandung ke Malang: ['Bandung', 'Cirebon', 'Yogyakarta', 'Surakarta', 'Surabaya', 'Malang']
Jarak: 854 km

Jakarta ke Yogyakarta: ['Jakarta', 'Cirebon', 'Yogyakarta']
Jarak: 537 km
C:\Storage\Documents\Telkom University\Semester 4\Dasar Kecerdasan Artifisial\praktikum\modul03\terbimbing\soal2.py:60: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might
be incorrect.
  plt.tight_layout()
```

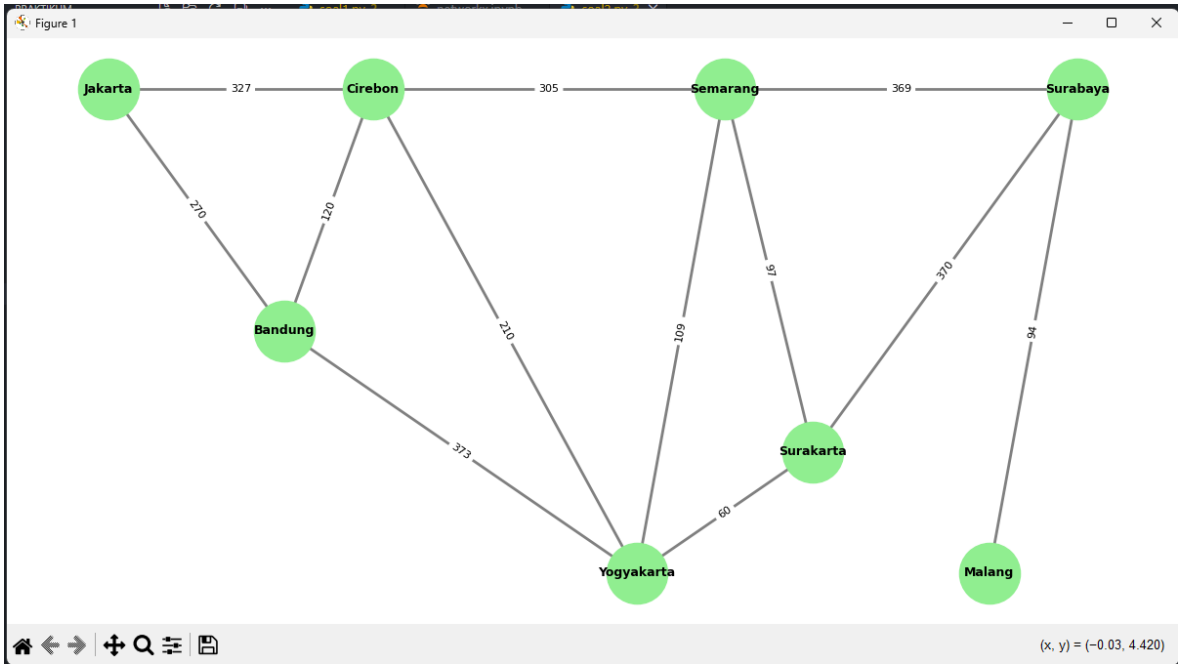
Figure 1
Jakarta
327
Cirebon
305
Semarang
369
Surabaya
270
120
210
109
97
370
94
Bandung
373
Surakarta
60
Yogyakarta
Malang
(x, y) = (−0.03, 4.420)